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Tyre and a method for manufacturing a tyre

Abstract

A tyre includes a body, a tread and an inner surface. Part of the inner surface has a noise-absorbing agent for absorbing tyre noise. The inner surface includes at least one spot free from noise-absorbing agent, provided with a module for wireless data transmission. The tyre includes a sensor arranged in the tread. The module is configured for reading data from the sensor. The module and the sensor are aligned to each other in the tyre. A method is for manufacturing such a tyre.

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Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATIONS
(1) The present application claims priority to Finnish Patent Application No. 20225885 filed on Oct. 3, 2022, which application is incorporated herein by reference in its entirety.
FIELD OF THE INVENTION
(2) The invention relates to a tyre for a vehicle. The invention relates to a vehicle tyre equipped with a sensor. The invention relates to a vehicle tyre having a low noise level. The invention relates to a method for manufacturing such a tyre.
BACKGROUND OF THE INVENTION
(3) It is known to provide a vehicle tyre with a sensor for measuring the properties of the tyre and/or the environment of the tyre, from the outside or the outer surface of the tyre. Such a sensor may communicate with a reader device. It is also known to provide noise-absorbing agent on the inside of the tyre, for reducing tyre noise.
(4) In a prior art solution, the reader device for the sensor can be arranged in the rigid part of the vehicle. Thus, noise-absorbing agent to be provided on the inside of the tyre does not hinder the attachment of the reader device. However, for data transmission, it is necessary to provide a reliable data transmission connection between the sensor and the reader device. A solution is also known in which the reader device is arranged in the tyre, within its carcass. It has been found that noise-absorbing agent hinders the fastening of the reader device.
BRIEF SUMMARY OF THE INVENTION
(5) It is an aim of the present invention to provide a solution for simultaneously securing reliable data transmission between the sensor and the reader device and the reliable attachment of the reader device in the tyre.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1***a* shows a tyre,
(2) FIG. **1***b* shows a quarter of a cross-section of a tyre,
(3) FIG. **2***a* shows a half of a cross-section of a tyre,
(4) FIG. **2***b* shows a half of a cross-section of a tyre according to an embodiment,
(5) FIG. **3***a* shows a part of a tyre according to an embodiment,
(6) FIG. **3***b* shows the area Mb of FIG. **3***a* in more detail,
(7) FIG. **4***a* shows a half of a cross-section of a tyre according to an embodiment,
(8) FIG. **4***b* shows a half of a cross-section of a tyre according to an embodiment,
(9) FIG. **5***a* shows a half of a cross-section of a tyre in a first step of manufacture of the tyre, and
(10) FIG. **5***b* shows a half of the cross-section of a tyre in a second step of manufacture of the tyre.
DETAILED DESCRIPTION OF THE INVENTION
(11) It is an aim of the present invention to provide a solution for simultaneously securing reliable

data transmission between a sensor in the tyre and a reader device, and reliable attachment of the reader device to a tyre comprising noise-absorbing agent. Further below, the reader device will be referred to by the term module. Noise-absorbing agent refers to a substance which is capable of damping noise and is or will be provided in the tyre for this purpose. Such a noise-absorbing agent is preferably ultralight substance as will be described hereinbelow.

(12) FIG. **1***a* shows a tyre **100**. The tyre **100** is suitable for use in a vehicle. The tyre **100** is preferably a pneumatic tyre, more preferably a tubeless pneumatic tyre. The tyre **100** may be, for example, a tubeless pneumatic tyre for a car, suitable for on-road conditions.

(13) The tyre **100** comprises a body (or carcass) **102** and a tread **112** which is suitable for rolling contact with a surface **900**. Furthermore, the tyre comprises an inner surface **111**. Part of the inner surface **111** is a surface opposite the tread **112**.

(14) FIG. **1***b* shows the structure of a typical tyre. FIG. **1***b* shows a quarter of the cross-section of a tyre. The structure of the tyre may be substantially mirror symmetrical with respect to the central axis Ax**1** shown in FIG. **1***b*; however, for example the tread pattern does not need to be mirror symmetrical. Preferably, the embodiments relate to a tyre **100** having such a structure. FIGS. **1***a* and **1***b* show the radial direction SR and the axial direction SAX of the tyre. The axial direction SAX is parallel to the rotation axis of the tyre. The radial direction SR is parallel to the radius of the tyre and extends away from the rotation axis.

(15) The tread **112** is typically formed by a tread block arrangement **121** (see FIG. **1***b*) comprising tread blocks **122** (see FIG. **2***a*) which delimit grooves in the tread **112** of the tyre **100**. Shoulder areas **135** of the tyre **100** are left at the edges of the tread bar arrangement **121**. As needed, the tread **112** may be provided with studs and/or sipes and/or mechanical indicators, such as running-in indicators.

(16) The tyre **100** has side surfaces **132**. The side surfaces **132** connect a bead area **134** to the tread **112**. The side surface **132** may be provided with various markings which indicate the size of the tyre, the speed category of the tyre, the intended use of the tyre (winter/summer), the manufacturer of the tyre, and/or the brand of the tyre. A cable **129** is provided in the bead area. The function of the cable **129** and the bead area **134** is to fit the tyre **100** on a rim. For stabilizing the tyre **100**, it comprises a first ply cord **127** and a first metal belt **125**. In an embodiment, the tyre also comprises a textile belt **123**. In a preferred embodiment, the tyre **100** comprises a first ply cord **127**, a second ply cord **126**, a first metal belt **125**, a second metal belt **124**, and a textile belt **123**.

(17) The first metal belt **124** or metal belts **124**, **125** are preferably steel belts. The first ply cord **127** or ply cords **126**, **127** preferably comprise a fibrous substance, such as fabric, glass fibre, aramid fibre, or carbon fibre. The textile belt **123** comprises a fibrous substance in textile form. Preferably, the textile belt **123** comprises fibrous polymer. Particularly preferably, the textile belt **123** comprises fibrous polyamide in textile form, for example aromatic polyamide in textile form. Materials comprising fibrous polyamides include, for example, nylon, aramid, and Cordura®.

(18) At least the first ply cord **127** and the first metal belt **125** together constitute a reinforcement **130** for the tyre. Preferably, the reinforcement **130** also comprises a textile belt **123**. More preferably, the reinforcement **130** further comprises a textile belt **123**, a second ply cord **126** and a second metal belt **124**.

(19) Furthermore, the tyre typically comprises an inner liner **131**. The English term ‘inner liner’ is commonly known in the field. A Finnish translation less frequently used for this term corresponds to ‘inner rubber liner’, but the established term ‘inner liner’ will be used in this specification. The function of the inner liner **131** (i.e. inner rubber liner **131**) is to prevent or decelerate the diffusion of air or other gas through the tyre **100**, in other words, to improve the air (or other gas) tightness of the tyre **100**. The inner liner **131** is thus arranged to improve the tightness of the tyre **100**. Most typically, the inner liner **131** comprises or is made of butyl rubber. Most preferably, the inner liner **131** comprises or is made of halobutyl rubber, such as bromobutyl rubber and/or chlorobutyl rubber, because such a material has good air tightness.

(20) Furthermore, noise-absorbing agent **180** is provided on the inside of the tyre **100**. The function of the noise-absorbing agent **180** is to reduce tyre noise. Thus, part of the inner surface **111** of the tyre **100** is consists of noise-absorbing agent **180** suitable for reducing tyre noise. In this description, the term 'inside' refers to that side of the tyre which is closer to the rotation axis of the tyre **100**, in the radial direction SR, than the tread **112**.
(21) Referring to FIG. **2***a*, the tyre **100** comprises a sensor **210** arranged in the tread **112**. More precisely, the sensor **210** is arranged in a tread block **122** in the tread block arrangement **121** of the tyre **100**. The sensor **210** is arranged in the tread **112** for measuring the tread **112** of the tyre and/or the environment outside the tyre. In an embodiment, part of the sensor **210** constitutes a part of the tread **112** of the tyre. Thus, the sensor **210** is also in contact with the environment, for example, via a blind hole **212** (see FIGS. **5***a* and **5***b*), and not totally embedded in the tyre material. The sensor **210** is preferably arranged to indicate at least the wear of the tread **112** of the tyre. In addition to the wear of the tread **112** of the tyre, the sensor **210** may indicate ambient moisture and/or temperature and/or tyre friction and/or acceleration of the sensor **210**. Furthermore, in addition to the wear of the tread **112** of the tyre, the sensor **210** may also indicate forces exerted on the tyre, whereby it may comprise, for example, an element for measuring strain and/or tension. Alternatively, it is possible that the sensor **210** is not arranged to indicate wear of the tread **112** of the tyre but is arranged to indicate another variable mentioned above in connection with the wear of the tread **112** of the tyre.
(22) The tyre **100** also comprises a module **220**. The module **220** is used, among other things, as a reader device for the sensor **210**. For this reason, the module **220** comprises means for reading information from the sensor **210**. Correspondingly, the sensor **210** comprises means for transmitting information.
(23) In an embodiment, the module **220** comprises means for inductively reading information from the sensor **210**. Correspondingly, the sensor **210** thus comprises means for inductively transmitting information. In an embodiment, the module **220** comprises a first inductive component, such as a coil, and the sensor **210** comprises a second inductive component, such as a coil, so that the first and second inductive components are arranged to be inductively connected to each other.
(24) Furthermore, the module **220** may comprise a sensor arrangement. The sensor arrangement of the module **220** may be arranged to measure at least one of the following: pressure, acceleration, and temperature.
(25) For improving the data transmission between the sensor **210** and the module **220**, the sensor **210** and the module **220** are aligned to each other. This alignment will be discussed further below. In an embodiment, the sensor **210** and the module **220** are aligned. However, the information obtained from the sensor **210** needs to be transmitted to a user and/or a server as well. Therefore, the module **220** also comprises means for wireless transmission of information. In an embodiment, the module **220** comprises an antenna for wireless transmission of information. In an embodiment, the module **220** comprises a power source for driving the module. Said power source may be an accumulator, a battery, a capacitor, or a converter (i.e. harvester) for converting (i.e. harvesting) other forms of energy, such as mechanical energy, to electric current; or a combination of these. Such a converter is also called an interceptor or a harvester, and the operation of the converter may be called scavenging of electric energy or harvesting of electric energy. In addition to the converter, i.e. harvester, it is possible to use a unit for storing electric energy, such as a battery or a capacitor. In an embodiment, the sensor **210** is arranged to generate the electric energy needed for its operation by means of said second inductive component included in the sensor **210**.
(26) The term 'tyre body **102**' has been presented above. The body **102** of the tyre refers to such a part of the tyre to which the tread **112** and the noise-absorbing agent **180** are or have been attached. In other words, the tyre body **102** does not comprise the noise-absorbing agent **180**, the sensor **210**, nor the module **220**. The body **102** does not need to comprise the tread block arrangement **121** of the tyre **100**.

(27) With reference to FIG. **2***b*, part of the inner surface **111** of the tyre **100** is formed by the noise-absorbing agent **180**. For fastening the module **200** to the tyre, the module **220** is arranged in a spot **182** free from noise-absorbing agent **180** in the tyre, either on or in the inner surface **111** of the tyre **100**. Thus, the inner surface **111** comprises at least one spot **182** which is free from noise-absorbing agent **180** and in which the module **220** is arranged. However, it is noted that there is not necessarily any space left free from noise-absorbing agent between the module **220** and the noise-absorbing agent **180**. In any case, there is no noise-absorbing agent at the very spot of the module **220**. Moreover, there is no noise-absorbing agent **180** left between the module **220** and the tyre body **102** on the inside of the tyre **100**, seen in the radial direction SR from the body **102**. Particularly preferably, the noise-absorbing agent **180** surrounds the module **220** in all directions of the inner surface **111** defined by the location **221** of the module (see FIGS. **3***a* and **3***b*). In other words, particularly preferably, the noise-absorbing agent **180** surrounds the module **220** in all directions perpendicular to the direction of the thickness of the tyre body **102** at the location of the module **220**. If the module **220** is aligned with the sensor **210** in the tread **112**, the noise-absorbing agent **180** preferably surrounds the module **220** in all directions perpendicular to the radial direction SR of the tyre.
(28) As will be described further in connection with the method below, such a spot **182** free from noise-absorbing agent can be arranged, for example, by removing noise-absorbing agent **180** from this spot or by providing noise-absorbing agent **180** at only other locations on the inner surface but not on the spot **182** to be free from noise-absorbing agent. As will be described further below, it is not necessary to provide noise-absorbing agent **180** on the whole inside of the tyre **100** either.
(29) The tyre **100** is vulcanized, i.e. hardened, during its manufacture.
(30) Such a tyre **100** is made, for example, by vulcanizing a tyre (i.e. an unvulcanized tyre), whereby the tyre **100** (vulcanized tyre as well as unvulcanized tyre) comprises said tread **112** suitable for rolling contact with a surface **900**. The tyre **100** (vulcanized tyre as well as unvulcanized tyre) further comprises an inner surface **111** as described above.
(31) Referring to FIG. **5***a*, a blind hole **212** for a sensor **210** is provided at a location **211** for the sensor in the tread **112** before, during or after the vulcanization in the method. Before vulcanization, the blind hole **212** can be formed by pressing with a suitable pin. During vulcanization, the blind hole **212** can be formed by a pin arranged in the tyre mould. When said pin is removed from the (vulcanized) tyre **100**, the blind hole **212** for the sensor is left in the tyre. After vulcanization, the blind hole **212** can be formed, for example, by machining, such as drilling, milling or cutting (e.g. by means of a laser), or burning (e.g. by means of a laser). Thus, it is not necessary to make a blind hole for the sensor during vulcanization.
(32) In the method, noise-absorbing agent **180** suitable for reducing tyre noise is provided on the inside of the tyre (vulcanized or unvulcanized) so that the inner surface **111** of the tyre is free from noise-absorbing agent at the location **221** for the module on the inside of the tyre. Preferably, the method comprises providing noise-absorbing agent **180** on the inside of the vulcanized tyre as described above; that is, the noise-absorbing agent **180** is arranged in the tyre **100** first after vulcanization.
(33) FIGS. **5***a* and **5***b* show an adhesive band or tape **190** for attaching noise-absorbing agent **180** (cf. also FIG. **4***b*). However, such a tape **190** is not necessarily needed.
(34) The method comprises aligning the location **211** for the sensor and the location for the module **221**. In an embodiment, the location **211** for the sensor is aligned with the location **221** for the module, or the location **221** for the module is aligned with the location **211** for the sensor. As will be described below, the alignment of both locations **221**, **211** can be performed with respect to a reference point, which reference point is neither of said locations **211**, **221**. The reference point may be, for example, a marking. Thus, after the installation of the module **220** and the sensor **210**, the module **220** and the sensor **210** are aligned to each other in the tyre **100**. Preferably, the location **211** for the sensor and the location **221** for the module are aligned to each other so that the location

211 for the sensor and the location **221** for the module are aligned.
(35) Furthermore, the method comprises installing the sensor **210** in the blind hole **212**. FIG. **5***b* illustrates a situation after the installation of the sensor **210** in the blind hole **212** (cf. also FIG. **5***a*). In FIG. **5***b*, the module **220** is not yet attached. The method comprises attaching the module **220** to the location **221** for the module. When the module is attached to the location **221** for the module shown in FIG. **5***b*, a tyre **100** substantially as shown in FIG. **4***b* is obtained. With respect to the adhesive bands or tapes, FIGS. **5***b* and **4***b* do not correspond to each other, but it is obvious that the tyre could be provided with similar adhesive bands or tapes **190** as in FIG. **4***b* already before attaching the module **220** (cf. FIG. **5***b*).
(36) The module **220** can be attached to the location **221** for the module by, for example, adhesive or tape, as will be described below. The sensor **210** can be installed in the blind hole **212** in the same way as a stud is installed in a winter tyre, for improving road holding on ice. What has been said above on the data transmission between the module **220** and the sensor **210**, and the respective means in connection with the tyre **100**, applies to the method as well. What has been said above on the energy used by the module **220** and the sensor **210** in connection with the tyre **100**, applies to the method as well. What has been said above on the means of the module **220** for transmitting data and/or for reading data from the sensor **210** in connection with the tyre **100**, applies to the method as well.
(37) In a preferred embodiment, noise-absorbing agent **180** is first, but after vulcanization, provided on the inner surface **111** of the tyre, including the location **221** for the module. In this embodiment, part of the noise-absorbing agent is removed from the inner surface **111** of the tyre, whereby the location **221** for the module is formed so that the inner surface **111** is free from noise-absorbing agent at the location **221** for the module.
(38) With reference to FIG. **4***a*, regarding the tyre, the sensor **210** and the module **220** are preferably aligned to each other so that a straight line SN parallel to the normal of the tread **112** of the tyre at the location of the sensor **210** passes through the module **220** and the sensor **210**. Alternatively or in addition, a straight line parallel to the normal of the inner surface **111** at the location of the module **220** passes through the module **220** and the sensor **210**. Furthermore, the module **220** and the sensor **210** are on the same side of the axis of rotation of the tyre. In other words, the distance between the module **220** and the sensor **210** is not greater than the thickness Tt of the tyre. In this context, the thickness Tt of the tyre refers to the thickness measured at said straight line, and the thickness Tt is left between the tread **112** and the inner surface **111**. In practice, the distance between the module **220** and the sensor **210** is slightly smaller than the thickness Tt, because the sensor **210** is installed in the blind hole in the tread **112** of the tyre. In such an embodiment, the distance between the sensor **210** and the module **220** is short, and data transmission is thus reliable.
(39) Correspondingly, in the method, the location **211** for the sensor and the location **221** for the module are preferably aligned to each other so that the straight line SN which is parallel to a normal of the tread **112** of the tyre at the location **211** for the sensor, passes through the location **221** for the module and the location **211** for the sensor. Alternatively or in addition, said alignment is such that a straight line parallel to the normal of the inner surface **111** of the tyre at the location **221** for the module passes through the location **221** for the module and the location **211** for the sensor.
(40) Furthermore, the location **221** for the module and the location **211** for the sensor are on the same side of the axis of rotation of the tyre. In other words, the distance between the location **221** for the module and the location **211** for the sensor is equal to or smaller than the thickness Tt of the tyre.
(41) In an embodiment of the method, noise-absorbing agent **180** is provided in locations other than the location **221** for the module on the inside of the tyre. The noise-absorbing agent **180** may be, for example, a sheet-like piece provided with a hole. Said hole can be provided at the location

221 for the module on the inner surface **111** of the tyre. For example, noise-absorbing agent **180** in hardenable fluid form can be extruded onto the inner surface **111** of the tyre, in locations other than the location **221** for the module. However, in such a case it has turned out to be difficult to align the location **221** for the module (i.e. the spot **182** to be free from noise-absorbing agent) with the location for the sensor.

(42) Therefore, in an embodiment of the method, noise-absorbing agent **180** is also provided at the location **221** for the module on the inside of the tyre, and after that, the noise-absorbing agent **180** is removed from the location **221** for the module. The noise-absorbing agent **180** may be, for example, a sheet-like solid piece provided on the inside of the tyre **100**. For example, noise-absorbing agent **180** in hardenable fluid form can be extruded onto the inner surface **111** of the tyre and removed from the location **221** for the module.

(43) In an embodiment of the method, a blind hole **212** is arranged at the location **211** for the sensor in the tread **112** before providing noise-absorbing agent **180** on the inside of the tyre **100**. This facilitates the making of the blind hole **212**. For example, the blind hole **212** can be made during vulcanization, and the noise-absorbing agent **180** can be provided not until in the vulcanized tyre. Alternatively, the blind hole **212** can be made in the tyre after vulcanization but before the installation of the noise-absorbing agent **180**.

(44) Preferably in this embodiment, the sensor **210** is also installed in the blind hole **212** before providing the noise-absorbing agent **180** on the inside of the tyre **100**. This facilitates the processing of the tyre as well. It is possible to provide a vulcanized tyre with the blind hole **212** not until after the installation of the noise-absorbing agent **180**, but in such a case it is typically more difficult to process the tyre when making the blind hole **212**.

(45) Furthermore, it is possible to provide the tyre, after its vulcanization, with the noise-absorbing agent **180** first and only then to arrange the blind hole **212** at the location for the sensor **211** in the tread **112**. In such a solution, to preserve the properties of the noise-absorbing agent, it may be preferable to inflate the tyre for the time of providing the tread **112** with the blind hole **212** and possibly also for the time of installing the sensor **210** in the blind hole. For these reasons, the tread **112** is preferably provided with the blind hole **212** at the location **211** for the sensor before providing the noise-absorbing agent **180** on the inside of the tyre **100**.

(46) Preferably, the blind hole **212** is machined at the location **211** for the sensor in the tread **112** after vulcanization, or the blind hole **212** is arranged in the tread **112** during vulcanization; and the noise-absorbing agent is installed after this. More preferably, the blind hole **212** is machined at the location **211** for the sensor in the tread **112** after vulcanization, and the noise-absorbing agent is installed in the tyre after this. This makes it possible to install a sensor **210** in only some of the tyres manufactured on a production line. Thus, two types of tyres can be made on a single production line, which increases the product range with low production costs.

(47) As is well known, machining in general can be implemented by a high-energy method or a low-energy method. Correspondingly, the machining of said blind hole can be implemented by a low-energy machining method or a high-energy machining method. Suitable machining methods include, in particular, drilling, milling, burning, and cutting, of which one or more methods can be applied in the machining of the blind hole **212**. Drilling and milling are low-energy methods. Burning and cutting can be implemented by using, for example, a laser, whereby the machining is carried out by a high-energy method. Preferably, the bottom hole **212** is machined by drilling.

(48) The alignment of the location **211** for the sensor and the location **221** for the module can be facilitated by means of a marking or markings. For example, the tyre **100** can be provided with a first marking **223** to indicate the location **221** for the module (see FIG. **2***b*), before installing the module **220**. Alternatively or in addition, the tyre **100** can be provided with a second marking **213** to indicate the location **211** for the sensor (see FIG. **2***b*), before making the blind hole **212**.

(49) For these reasons, in an embodiment of the method, the inner surface **111** of the tyre **100** is provided with a first marking **223** to indicate the location **221** for the module, and/or the outer

surface, for example the tread **112**, of the tyre is provided with a second marking **213** to indicate the location **211** for the sensor. For the first marking **223** in particular, it is noted that the first marking **223** is preferably made at such a location where the first marking **223** is visible also after the provision of the noise-absorbing agent **180**. In other words, the method comprises providing the noise-absorbing agent **180** on the inside of the tyre **100** without covering the first marking **223** which shows the location **221** for the module **220**. This makes it possible to use the first marking **223** for removing some of the noise-absorbing agent **180** later on, to form a spot **182** free from noise-absorbing agent at the location **221** for the module. The markings **211**, **223** do not need to be removed. Therefore, the finished tyre **110** may also comprise at least one of the following: the first marking **223**, showing the location **221** for the module, and the second marking **213**, showing the location **211** for the sensor.

(50) As presented above, preferably the blind hole **212** is made before providing the noise-absorbing agent **180**. Naturally, the blind hole **212** as such indicates the location **211** for the sensor. Therefore, preferably after the drilling or providing of the blind hole **212**, noise-absorbing agent **180** is provided on the inside of the tyre, and the first marking **223** is applied on the inner surface **111** of the tyre, to indicate the location **221** for the module. More preferably, said first marking **223** is applied before providing the noise-absorbing agent **180** and after making the blind hole **212**. Most preferably, the sensor **210** is installed in the blind hole **212**, too, before noise-absorbing agent is provided on the inside of the tyre **100**. The first marking **223** is applied on the inner surface **111** of the tyre preferably with a dye, such as paint or ink. In this context, dye also refers to ultraviolet dyes, i.e. substances which are particularly well visible under UV light. Preferably, the colour of the dye is not the same as the colour of the inner surface **111** of the tyre. Preferably, the colour of the first marking **223** is different from black.

(51) A first marking **223** indicating such a method may be left on the tyre **100**. Therefore, one embodiment of the tyre **100** comprises a first marking **223** on the inner surface **111**, indicating the location **221** for the module. The first marking **223** may comprise a dye as described above.

(52) In an embodiment, the tyre has a secondary marking **105** (see FIG. **1***a*) which primarily serves a different purpose, such as indicating the tyre size (e.g. “215/55R16”) or indicating the brand name of the tyre (e.g. “Hakkapeliitta”) or indicating the use environment of the tyre (for example, “M/S”), and such a secondary marking **105** is utilized in determining the location **221** for the module **220** and, optionally, also in determining the location **211** for the sensor **210**. Such a secondary marking **105** may be applied in the tyre, for example, during vulcanization in a mould. The secondary marking **105** can be detected from the tyre and used for aligning the location **211** for the sensor with the location **221** for the module. For example, the location **211** for the sensor and the location **221** for the module may have specific coordinates with respect to such a secondary marking **105**. Said coordinates are preferably such that the location **221** for the sensor and the location **221** for the module are aligned as described above. For example, the sensor **210** can be installed in a tread block and, after installing, the location **211** (i.e. coordinates) of the sensor can be detected in relation to the secondary marking **105**. After this, the location for the module can be determined by means of said coordinates and said secondary marking **105**.

(53) In an embodiment, the secondary marking **105** is detected automatically, for example by machine vision. Although the secondary marking **105** is shown on the side surface of the tyre in FIG. **1***a*, such a secondary marking **105** may be arranged on the tread. Thus, the same marking may be said secondary marking **105** and said second marking **213**. For example, the tread may be provided with a marking indicating a risk of aquaplaning or wear. When the location **211** for the sensor **210** is known, it is possible to align the location **211** for the sensor and the location **221** for the module.

(54) In an embodiment, after installing the sensor **210** in the blind hole, the location **211** of the sensor **210** is detected. The location **211** of the sensor can be detected, for example, inductively. Upon detecting the location **211** of the sensor, the strength of the signal received from the sensor

indicates the location of the sensor. The location **211** of the sensor can be detected, for example, from the side of the inner surface **111** of the tyre. After this, the location **221** for the module **220** can be aligned with the location **211** of the sensor **210** detected in this way. In this embodiment, the first or second marking **213**, **223** or the secondary marking **105** is not necessarily needed.

(55) In an embodiment, the module **220** is attached to the tyre **100** by means of adhesive or tape. If the module **220** is attached to the tyre **100** by means of adhesive or tape, then, in the finished tyre **100**, the module **220** is attached to the body **102** of the tyre **100** by means of adhesive or tape. For clarity, it is noted that during the manufacture, the module is not necessarily part of the tyre but the module is attached to the tyre **100** (for example, to the body of the tyre). When the tyre is finished, the module is part of the tyre, the module being fastened to the body **102** of the tyre.

(56) The module **220** does not need to be attached directly to the tyre **100**. For example, EP 3 543 042 presents how the module can be installed in a receptacle. Such a receptacle can be attached to the tyre **100** by means of adhesive or tape, and the module **220** can be installed in the receptacle, whereby the module **220** is fastened to the tyre **100** by means of adhesive or tape and also by means of the receptacle. In this case, too, the module **220** is attached to the tyre **100** by at least adhesive or tape. Correspondingly, in a tyre comprising the module **220**, the module **220** is attached to the body **102** of the tyre **100** by at least adhesive or tape.

(57) Particularly advantageously, double-sided tape is used for attaching the module. Thus, removing the noise-absorbing agent **180** becomes easier. Double-sided tape refers to a structure in which adhesive is provided on a first side of a base material and adhesive is provided on a second side of the base material. Moreover, a release film is provided on the adhesive on the second side of the base material. Furthermore, a release film may be provided on the adhesive on the first side of the base material.

(58) When double-sided tape is used, the module **200** is preferably attached to the tyre **100** by said double-sided tape **230** (see e.g. FIGS. **5***b* and **4***b*) so that the first side of the double-sided tape **230** is attached to the inner surface **111**, at the location **221** for the module, after vulcanization of the tyre. If a release film is provided on the adhesive on the first side of the base material, this release film is removed to expose the adhesive, to make the tape adhere to the inner surface **111** of the tyre at the location for the module.

(59) After attaching the double-sided tape **230** to the tyre **100**, noise-absorbing agent **180** is provided on the inside of the tyre, suitable for reducing tyre noise, so that the noise-absorbing agent **180** also covers the double-sided tape **230**. To expose the spot **182** to be free from noise-absorbing agent, noise-absorbing agent **180** is removed from the location **221** for the module. Therefore, in an embodiment, noise-absorbing agent **180** is removed from the area defined by the double-sided tape **230**. The release film of the double-sided tape facilitates the removal of the noise-absorbing agent **180**, because the release film of the tape **230** can be removed simultaneously with removing the noise-absorbing agent **180**. In other words, the noise-absorbing agent **180** can attach to the release film, which facilitates the removal of the noise-absorbing agent, compared with a situation in which the noise-absorbing agent **180** were attached to the tyre body.

(60) Preferably, the noise-absorbing agent **180** is removed from the area defined by the double-sided tape **230** so that after providing the noise-absorbing agent **180**, an incision is cut in the noise-absorbing agent **180** at the location of the double-sided tape **230**, and the noise-absorbing agent **180** is removed from the area defined by the incision.

(61) More preferably, in this embodiment, adhesive covered with a release film is provided on the second side of the double-sided tape **230** (opposite the first side of the double-sided tape **230**). Furthermore, the noise-absorbing agent **180** is removed from the area defined by the incision, and the release film is removed from the double-sided tape **230**. For example, the noise-absorbing agent **180** may be removed from the area defined by the incision by simultaneously removing the release film from the double-sided tape **230**. For example, the noise-absorbing agent **180** may attach to the release film of the double-sided tape **230**, whereby, upon removing noise-absorbing agent, the

release film is also removed and the adhesive of the tape **230** is exposed. Irrespective of the way of removing the release film, the adhesive on the second side of the double-sided tape can be used for attaching the module.

(62) When double-sided tape has been used for attaching the module **220**, the module **220** in the tyre **100** is fastened to the body **102** of the tyre **100** by at least double-sided tape. If the above-mentioned receptacle is used, such a receptacle can be attached to the tyre **100** (that is, it can be attached to the tyre body **102**) by double-sided tape **230**. If the module **220** is attached to the body **102** of the tyre **100** by at least double-sided tape, base material of the double-sided tape is arranged between the tyre body **102** and the module **220**. The structure of the double-sided tape was discussed in more detail above.

(63) In removing the noise-absorbing agent **180**, a tool, such as a laser, a saw, a drill, a mill, or a gripper, is used. For example, a laser, a saw, a drill, or a mill can cut an incision in the noise-absorbing agent, to be used for removing noise-absorbing agent. The cutting of the incision can be done by burning or by removing material in another way as described above in connection with the method. For example, a gripper can be used to rip or pull part of the noise-absorbing agent off the location **221** for the module. For example, a mill or a drill can be used to remove some of the noise-absorbing agent from the location **221** for the module. For example, a laser can be used to burn some of the noise-absorbing agent off the location **221** for the module. Noise-absorbing agent **180** can be removed by using an above-described device even if double-sided tape were not used for attaching the module **220**.

(64) In an embodiment, the device used for removing noise-absorbing agent **180** is aligned with the correct location for removing noise-absorbing agent **180** and/or for cutting an incision in the noise-absorbing agent **180**, for example, so that the location **221** for the module is aligned with the location **211** for the sensor. In an embodiment, the device used for removing noise-absorbing agent **180** is aligned by a marking **223** on the inner surface **111** of the tyre **100** to the correct location for removing noise-absorbing agent **180** and/or for cutting an incision in the noise-absorbing agent **180**. Naturally, when such a marking is used, the marking **223** is provided on the inner surface **111** of the tyre as described above. In an embodiment, the device used for removing noise-absorbing agent is aligned by means of a secondary marking **105** in the tyre **100** to the correct position for removing noise-absorbing agent **180** and/or for cutting an incision in the noise-absorbing agent **180**. In an embodiment, the location **211** for the sensor **210** is detected, and the device used for removing noise-absorbing agent **180** is aligned, by using the data on the position of the sensor **210** obtained in this way, to the correct location for removing noise-absorbing agent **180** and/or for cutting an incision in the noise-absorbing agent **180**. In these embodiments, the alignment can be performed by the user, or it can be automated. An automated arrangement, in particular, may further comprise a camera for detecting the first marking **223** or the secondary marking **105**. Thus, in an embodiment, the device used for removing noise-absorbing agent **180** is aligned, by means of the first marking **223** on the inner surface **111** of the tyre **100**, using a camera, to the correct location for removing noise-absorbing agent **180** and/or for cutting an incision in the noise-absorbing agent **180**. In an embodiment, the device used for removing noise-absorbing agent **180** is aligned by means of the secondary marking **105** in the tyre **100**, using a camera, to the correct location for removing noise-absorbing agent **180** and/or for cutting an incision in the noise-absorbing agent **180**.

(65) In an embodiment, an incision is cut in the noise-absorbing agent **180** at the location **221** for the module, by means of a device used for removing noise-absorbing agent **180**, such as a laser, a saw, a mill, or a drill, and the noise-absorbing agent **180** is removed from the area defined by the incision. This can be done, for example, when the noise-absorbing agent **180** at the location **221** for the module is not attached to the tyre **100** at all. This can be done, for example, by attaching the noise-absorbing agent **180** to the inside of the tyre by means of adhesive or tape elsewhere than at the location **221** for the module.

(66) In an embodiment, the noise-absorbing agent **180** is attached to the inner surface **111** of the tyre by tape or adhesive. In this embodiment, the noise-absorbing agent may be a solid flexible sheet, such as a sheet of polymer foam, which is attached by tape or adhesive.
(67) For the above mentioned reasons, in an embodiment, the noise-absorbing agent **180** is attached to the inner surface **111** of the tyre by tape or adhesive so that the location **221** for the module is a spot free from adhesive or tape between the noise-absorbing agent **180** and the tyre body **102**. Particularly between the noise-absorbing agent **180** and the tyre body **102**, there is no said adhesive or tape at the location **221** for the module outside the noise-absorbing agent **180** in the radial direction SR. In a tyre **100** manufactured in this way, the noise-absorbing agent **180** is attached to the tyre body **102** by adhesive or tape.
(68) More preferably, the noise-absorbing agent **180** is attached to the inside of the tyre by at least one adhesive band or tape, whereby the location **221** for the module is a spot free from said adhesive band or tape between the noise-absorbing agent **180** and the tyre body **102**. In a tyre **100** manufactured in this way, the noise-absorbing agent **180** is attached to the tyre body **102** by adhesive band or tape **190**. Furthermore, in the method, the location **221** for the module is a spot free from said adhesive band or tape between the noise-absorbing agent **180** and the tyre body **102**, whereby the adhesive band or tape **190** in the corresponding tyre **100** does not extend all the way to the module **220**. With respect to the adhesive band or tape **190**, reference is made to FIG. **4***b*. As shown in the figure, in an embodiment, the adhesive band or tape **190** does not extend in the area **182** free for noise-absorbing agent. Thus, said band **190** does not extend to the module **220** either; that is, the adhesive band or tape **190** does not extend to the module **220**. The tape or adhesive band used (if used) in the attachment of the noise-absorbing agent **180** may be double-sided tape as described in connection with the module **220**.
(69) Before attaching, the noise-absorbing agent **180** may be, for example, solid material, such as polymer foam. Thus, a sheet of suitable size can be formed of the noise-absorbing agent **180** before it is installed on the inside of the tyre **100**, and it can be attached to the inside of the tyre **100** by, for example, said adhesive or tape. If the noise-absorbing agent **180** is provided on the tyre **100** in this way, a seam **185** is formed in the noise-absorbing agent (see FIG. **3***a*). Such a seam **185** extends from a first side to an opposite second side of the noise-absorbing agent **180** on the inner surface **111** of the tyre **100** (see FIG. **3***a*). The seam **185** may extend substantially in the axial direction SAX (see FIG. **1**), as shown in FIG. **3***a*. The seam **185** does not necessarily extend in the axial direction SAX. The seam **185** is not necessary straight either. The direction and the straightness of the seam **185** depend on, among other things, the shape of the sheet of noise-absorbing agent to be attached inside the tyre, which may be e.g. a rectangle or a parallelogram whose end sides (forming the seam **185**) are not necessary straight but may have e.g. a serrated or corrugated pattern. More than one seam **185** may be provided as well.
(70) Preferably, the module **220** does not cut said seam **185** in the noise-absorbing agent. This is because the noise-absorbing agent **180** is preferably attached to the tyre body **102** (or will be attached to the tyre) particularly close to the seam **185**, to secure the attachment of the noise-absorbing agent **180**. On the other hand, for the above presented reasons, a spot **182** free from noise-absorbing agent is provided at the module **220** on the inner surface **111** of the tyre **100**. These both aims can be achieved when the module **220** does not cut said seam **185** of the noise-absorbing agent. In other words, the seam **185** extends from the first side to the opposite second side of the noise-absorbing agent **180** on the inner surface **111** of the tyre **100** without intersecting the spot **182** free from noise-absorbing agent. Furthermore, the fact that the seam **185** extends without intersecting the spot **182** free from noise-absorbing agent allows the location for the sensor **210** to be selected more freely, because the module **220** is aligned with the sensor **210** in the tyre **100**.
(71) To secure better attachment of the noise-absorbing agent **180** to the tyre **100**, one embodiment comprises cleaning of the inner surface **111** of the tyre before providing the noise-absorbing agent **180**. More precisely, in one embodiment, the inner surface **111** of the tyre is cleaned after

vulcanization and before providing the noise-absorbing agent **180**. Cleaning improves the attachment both in the embodiment in which adhesive or tape is used for attaching the noise-absorbing agent, and in the embodiment in which the noise-absorbing agent as such adheres to the tyre (without adhesive or tape). For example, the inner surface **111** of the tyre, such as a vulcanized tyre, can be cleaned by laser.

(72) Cleaning may be necessary particularly because release chemicals may be used during vulcanization, to facilitate the removal of the vulcanized tyre from its manufacturing mould. Residues of such release chemicals may be left on the inner surface **111** of the tyre and thereby hamper the attachment of the noise-absorbing agent to the tyre.

(73) It is also possible to protect the inner surface **111** of the tyre during vulcanization, for example with a suitable film. In such a solution, instead of cleaning the inner surface, the protective film can be removed from the inner surface **111** of the tyre.

(74) As presented above, the tyre preferably comprises an inner liner **131** provided to improve the tightness of the tyre. In such an embodiment, before providing the noise-absorbing agent **180**, the inner liner **131** typically constitutes the inner surface **111** of the tyre. Thus, in an embodiment of the method, noise-absorbing agent **180** is provided on the inside of such a tyre **100** where an inner liner **131** constitutes the inner surface **111** before the noise-absorbing agent is provided in the tyre.

(75) Although, in an embodiment, noise-absorbing agent **180** is removed from such a tyre in the above described way, and the noise-absorbing agent **180** can be cut in connection with the removal, the inner liner **131** should not be damaged (e.g. punched) when cutting the noise-absorbing agent, to secure the operation of the tyre. Therefore, in an embodiment, the location **221** for the module is formed by removing part of the noise-absorbing agent **180** without making a hole in the inner liner; that is, in such a way that the inner liner **131** remains intact. What has been said on the material of the inner liner **131** in connection with the tyre also applies in connection with the method. Such a tyre **100** comprises an inner liner **131** provided to improve the tightness of the tyre **100**. At least part of the inner liner **131** is arranged between the noise-absorbing agent **180** and the outer surface of the tyre **100**. More precisely, at least part of the inner liner **131** is arranged in such a space between the noise-absorbing agent **180** and the outer surface of the tyre **100** that is outside the noise-absorbing agent **180** in the radial direction SR (see FIGS. **1***a* and **1***b*). As presented above, the inner liner **131** is unbroken, particularly in a tyre **100** having a module **220**.

(76) As presented above, the tyre preferably comprises a reinforcement **130**. Preferably, the reinforcement comprises at least one of the following: [A] a ply cord (**126**, **127**) comprising a fibrous substance, and [B] a metal belt, such as a steel belt. The reinforcement **130** is arranged between the tread **112** and the inner surface **111**. Preferably, the tread **112** is provided with said blind hole **212** at the location **211** for the sensor so that the blind hole **212** does not extend through the reinforcement **130**. In other words, the blind hole **212** is arranged at the location **211** for the sensor without punching the reinforcement **130**. Thus, the reinforcement **130** remains intact, although the blind hole **212** is formed in the tyre, whereby the reinforcement **130** reinforces the tyre **100** as desired. Such a tyre **100** comprises a reinforcement **130**, such as a ply cord (**126**, **127**) or a metal belt (**124**, **125**), between the tread **112** and the inner surface **111**. The tyre **100** also comprises a sensor **210** which does not extend through the reinforcement **130**. In particular, said reinforcement **130** is outside the inner surface **111**, in the radial direction SR (see FIGS. **1***a* and **1***b*).

(77) In an embodiment, the noise-absorbing agent **180** is ultralight material, such as foam (e.g. cellular foam) or other material of light weight. In this context, ultralight material refers to a material having a density lower than 10 kg/m.sup.3.

(78) Preferably, the noise-absorbing material **180** comprises polymer-based foam, i.e. polymer foam. In this context, polymer foam refers to dispersion of gas in a polymer matrix. A polymer foam consists of at least two phases: a solid polymer matrix and a gas phase. The gas phase may also be called blowing agent. Other solid phases may also be present in foams, for example in the form of bulking agents. Polymer foams may be, for example, expanded rubber or cellular elastomer

or spongy. The polymer matrix may be thermoplastic or thermosetting plastic. Physical and mechanical properties of polymer foam differ significantly from those of a corresponding solid polymer matrix material. Cellular plastic is an example of polymer foam.

(79) The matrix material for polymer foam can be selected from a plurality of polymers as needed. In an example according to the method, the matrix material for polymer foam is selected, or the matrix material for polymer foam in the tyre has been selected, from a group consisting of the following materials: ethylene vinyl acetate (EVA), polyethylene (PE), nitrile rubber (NBR), polychloroprene, neoprene, polyimide, polypropylene, polystyrene, polyurethane, polyvinyl chloride (PVC), and silicone.

(80) Examples of other substances which are ultralight with respect to their density include aerogels (such as silicon dioxide aerogel and carbon nanotube aerogel), other foams (such as metal foams), microlattices (such as metal microlattices), and aerographite.

(81) In an example, the noise-absorbing agent **180** is polymer foam. In an example, the noise-absorbing agent **180** is polymer foam and comprises polyurethane.

(82) In an embodiment, the thickness of the noise-absorbing agent **180**, measured in the radial direction SR of the tyre, is at least 10 mm. In an embodiment, the thickness of the noise-absorbing agent, measured in the radial direction SR of the tyre, is not greater than 75 mm. In an embodiment, the thickness of the noise-absorbing agent, measured in the radial direction SR of the tyre, is 20 mm to 60 mm, more preferably 30 mm to 50 mm. Such dimensions are generally applicable to foamy noise-absorbing agents and cellular foams as presented above.

(83) In an embodiment, the volume of the bulking agent **180** is at least 20% of the volume delimited by the body **102** of the tyre **100** and a rim suitable for the tyre.

Claims

1. A method for manufacturing a tyre, comprising: providing an unvulcanized tyre; vulcanizing the tyre, forming a blind hole for a sensor in the tyre before or during the vulcanizing the tyre, wherein the tyre comprises a tread, the blind hole for the sensor, and an inner surface, installing the sensor in the blind hole for the sensor, providing a noise-absorbing agent for absorbing tyre noise, on the inside of the vulcanized tyre so that: part of the inner surface comprises a location for a module, free from the noise-absorbing agent, attaching the module to the location for the module, wherein: the module comprises means for wireless data transmission, the module comprises means for reading data from the sensor inductively, and the sensor comprising means for inductive data transmissions to said module; the method comprising: detecting the blind hole for the sensor or the sensor, and aligning the location for the module by using location data obtained by the detection of the blind hole or the sensor so that a straight line which is parallel to a normal of the tread of the tyre at the location for the sensor passes through the location for the module and the location for the sensor, and/or a straight line which is parallel to the normal of the inner surface of the tyre at the location for the module passes through the location for the module and the location for the sensor, and after providing the noise-absorbing agent on the inside of the vulcanized tire, removing a portion of the noise-absorbing agent to form the location for the module, at which the inner surface of the tyre is free from the noise-absorbing agent.
2. The method according to claim 1, comprising: installing the sensor in the blind hole before providing the noise-absorbing agent on the inside of the tyre.
3. The method according to claim 1, comprising: attaching the module to the tyre by at least adhesive or tape.
4. The method according to claim 1, comprising: attaching the module to the tyre by a double-sided tape by: attaching a first side of the double-sided tape to the location for the module on the inner surface of the tyre after vulcanization of the tyre; providing a second side of the double-sided tape with an adhesive and a release film covering the adhesive; providing the noise-absorbing agent on

the inside of the tyre so that the noise-absorbing agent covers the double-sided tape; removing the noise-absorbing agent from an area defined by the double-sided tape by cutting an incision at the location of the double-sided tape in the noise-absorbing agent, removing the noise-absorbing agent from an area defined by the incision; and removing the release film from the double-sided tape.

5. The method according to claim 1, comprising: applying a tool comprising a laser, a saw, a drill, a mill, or a gripping device, for removing the noise-absorbing agent, provided on the inside of the vulcanized tyre, and/or for cutting an incision in the noise-absorbing agent provided on the inside of the vulcanized tyre, wherein the tool is aligned with the location of the module.

6. The method according to claim 1, comprising: attaching the noise-absorbing agent to the inner surface of the tyre by tape or adhesive, wherein: the location for the module is where no adhesive or tape for attaching said noise-absorbing agent is provided between the noise-absorbing agent and the body of the tyre.

7. The method according to claim 1, comprising: cleaning the inner surface of the tyre before providing the noise-absorbing agent, or removing a protective film from the inner surface of the tyre before providing the noise-absorbing agent.

8. The method according to claim 1, wherein: the tyre comprises an inner liner arranged to improve tightness of the tyre, and the location for the module is formed by removing part of the noise-absorbing agent without making a hole in the inner liner.

9. The method according to claim 1, wherein: the tyre comprises a reinforcement comprising a ply cord and/or a metal belt, between the tread and the inner surface, wherein the method comprises: providing the tread with said blind hole at the location for the sensor without punching the reinforcement.

10. The method of the claim 1, wherein: a density of the noise-absorbing agent is lower than 10 kg/m.sup.3 or the noise-absorbing agent comprises foam, and/or a thickness of the noise-absorbing agent is at least 10 mm.

11. A method for manufacturing a tyre, comprising: vulcanizing the tyre, wherein the tyre comprises a tread and an inner surface; providing a noise-absorbing agent for absorbing tyre noise, on an inside of the tyre so that part of the inner surface comprises a location for a module, free from the noise-absorbing agent; attaching the module to the location for the module; wherein:
the module comprises means for wireless data transmission, the module comprises means for reading data from a sensor inductively; providing a blind hole for the sensor at a location for the sensor in the tread before, during, or after vulcanization; aligning the location for the sensor and the location for the module to each other so that a straight line which is parallel to a normal of the tread of the tyre at the location for the sensor passes through the location for the module and the location for the sensor, and/or a straight line which is parallel to the normal of the inner surface of the tyre at the location for the module passes through the location for the module and the location for the sensor; installing the sensor in said blind hole, the sensor comprising means for inductive data transmission to said module; attaching the module to the tyre by a double-sided tape by: attaching a first side of the double-sided tape to the location for the module on the inner surface of the tyre after vulcanization of the tyre; providing a second side of the double-sided tape with an adhesive and a release film covering said adhesive; providing the noise-absorbing agent on the inside of the tyre so that the noise-absorbing agent covers the double-sided tape; removing the noise-absorbing agent from an area defined by the double-sided tape by cutting an incision at a location of the double-sided tape in the noise-absorbing agent; removing the noise-absorbing agent from the area defined by the incision; and removing the release film from the double-sided tape.